Measuring the Cooling of Warm Water

Target document A | Synthetic laboratory report

1. Aim

The aim was to observe how the temperature of warm water changes as it cools in a room.

2. Materials

A beaker, warm water, a thermometer, a stopwatch, and a notebook were used.

3. Method

Warm water was poured into the beaker. Its temperature was read at the start and then every two minutes for ten minutes. The thermometer was kept in the water without touching the sides of the beaker.

4. Observations

The recorded temperatures were: start 70 degrees C; 2 minutes 61 degrees C; 4 minutes 54 degrees C; 6 minutes 48 degrees C; 8 minutes 44 degrees C; 10 minutes 41 degrees C.

5. Discussion

The water temperature decreased over time. The decrease became smaller toward the end of the observation period, which is consistent with cooling toward room temperature.

6. Conclusion

The water cooled during the ten-minute observation, so the experiment demonstrated a decrease in water temperature over time.

No specific hazards or safety precautions are discussed in this report.

Designed test intent: aim, materials, procedure, observations, analysis, and result are represented; the safety point is deliberately omitted.